



Cloud Storage Solution Design



Foreword

This lesson gives an overview of Huawei Cloud storage solutions, including basic storage services (OBS, SFS, and EVS) and five principles of solution design.



Objectives

Upon completion of this course, you will understand:

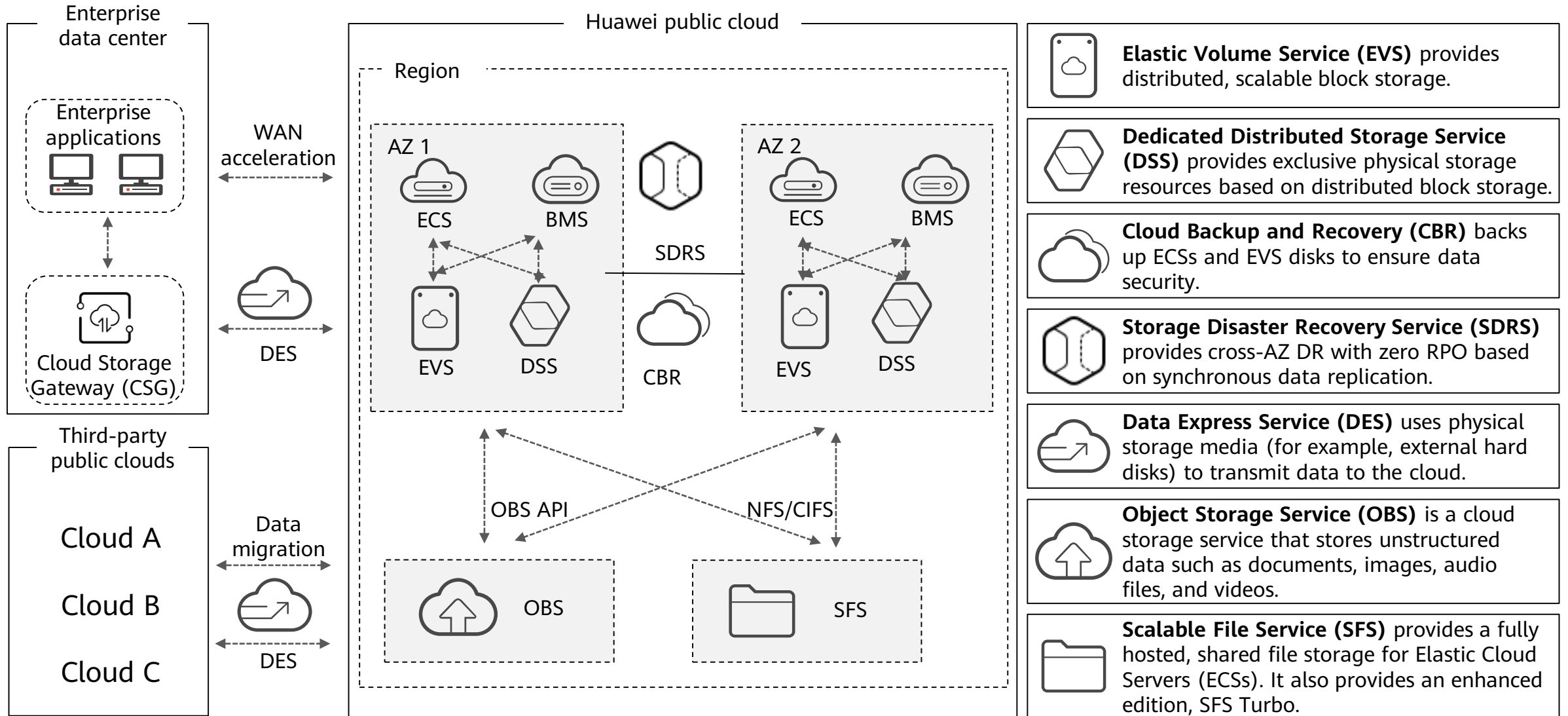
- Functions and features of OBS
- Specifications selection of SFS and EVS
- Five principles for designing storage solutions



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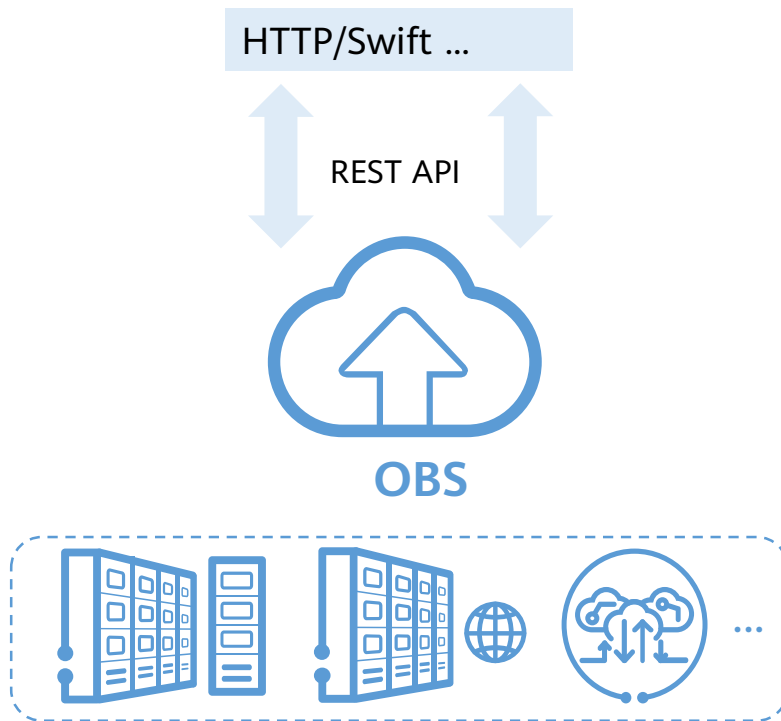
- 1. Storage Service Overview**
2. Storage Solution Design Principles
3. Cloud Compute and Storage Service Practices

Overview of Huawei Cloud Storage Services



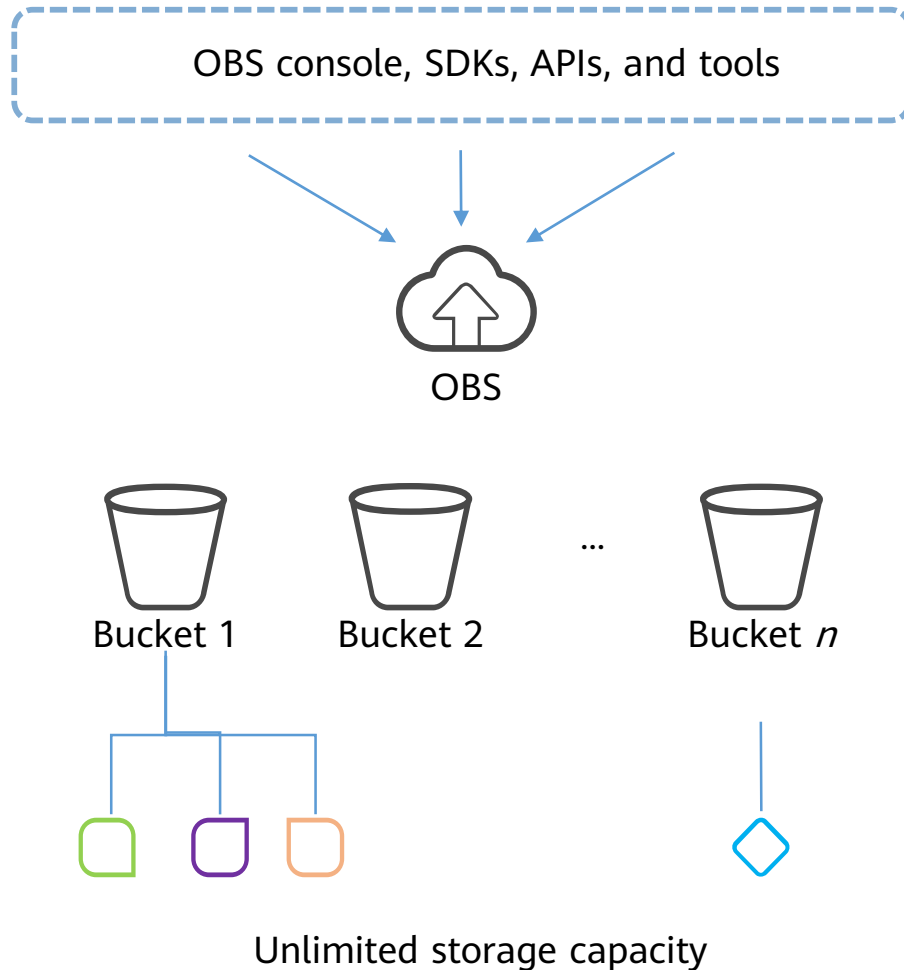
OBS

OBS is a secure, highly reliable object storage service that allows you to inexpensively store massive data.



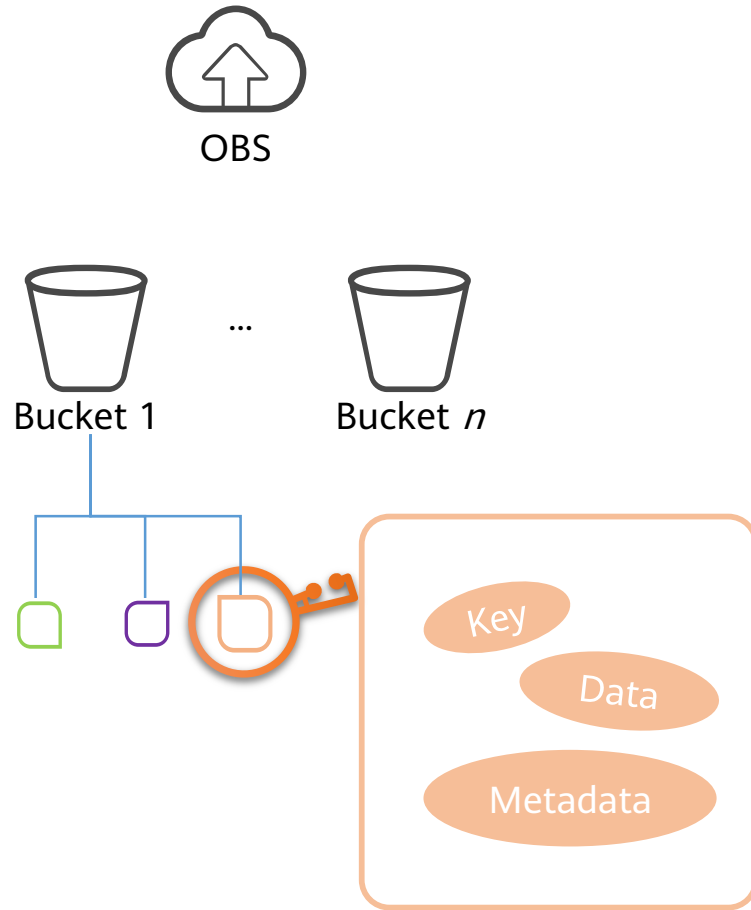
- Fully managed
- Accessible over the Internet
- Unlimited storage (allowing for a single object of up to 48 TB)
- Reliability of 99.9999999999%
- Event notifications
- Various, cost-effective solutions

Getting Started with OBS



- You can use the console, SDKs, and various tools to access OBS.
- To store objects in OBS, you need to create a bucket first.
 - ▣ A bucket is the management unit of OBS.
- You can store any number of objects in a bucket.
 - ▣ An object is a logical unit of storage in OBS.
 - ▣ You can store files of various types in OBS, such as documents, images, and audio files.
- You can manage a bucket by:
 - ▣ Storing objects with the same attribute or associated with the same service in the same bucket.
 - ▣ Assigning unified management policies to objects in the same bucket.

Identifying an Object



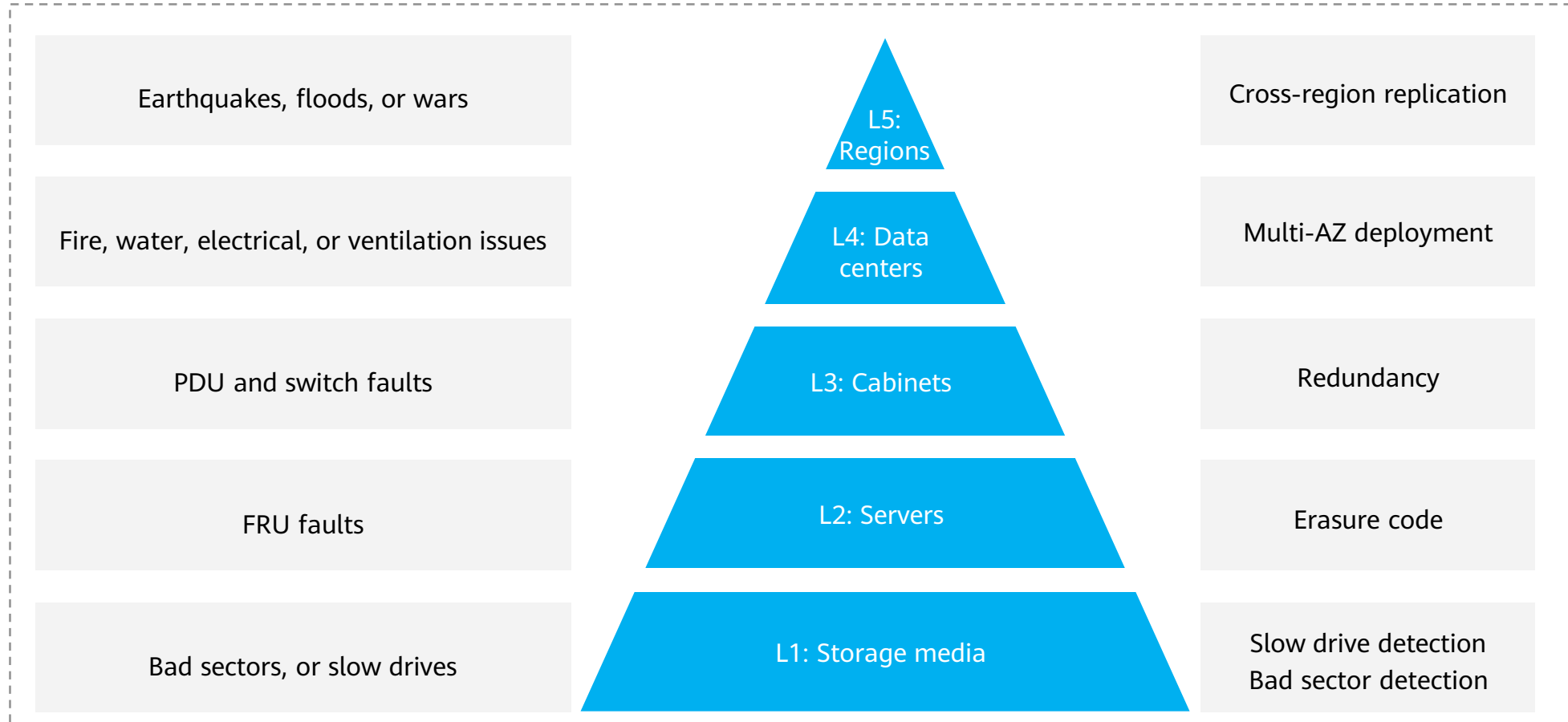
An object consists of:

- ❑ A key: Object name. Each object in a bucket must have a unique key.
- ❑ Metadata: System and user-defined metadata in the form of key-value pairs, used to describe the object.
- ❑ Data: Object content.

An object key uniquely identifies an object in the bucket.

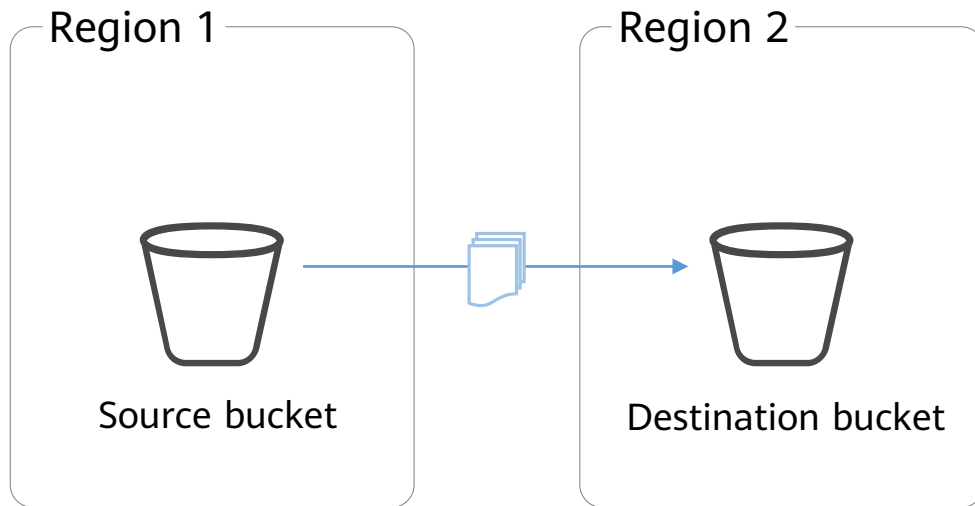
Reliability and Availability of OBS

Reliability of 99.9999999999%, availability of 99.995%



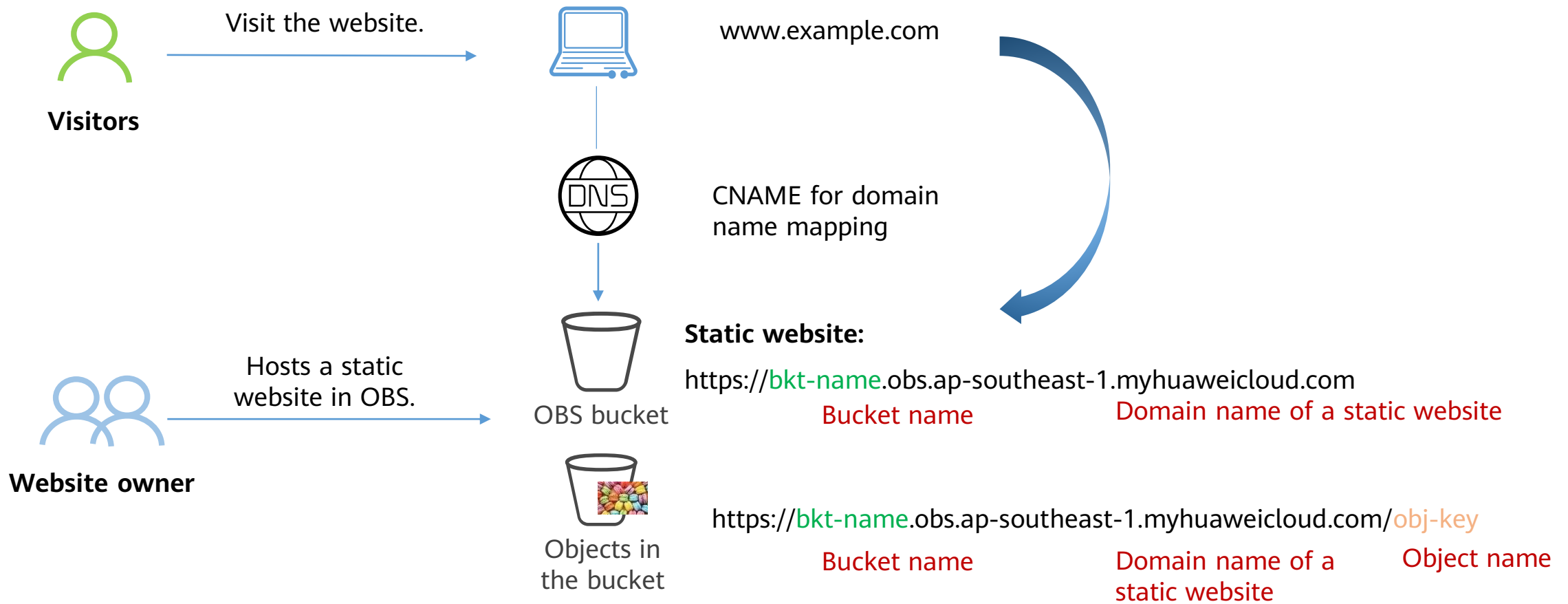
Cross-Region Replication of OBS

You can configure a cross-region replication rule to enable automatic, asynchronous data replication across regions.



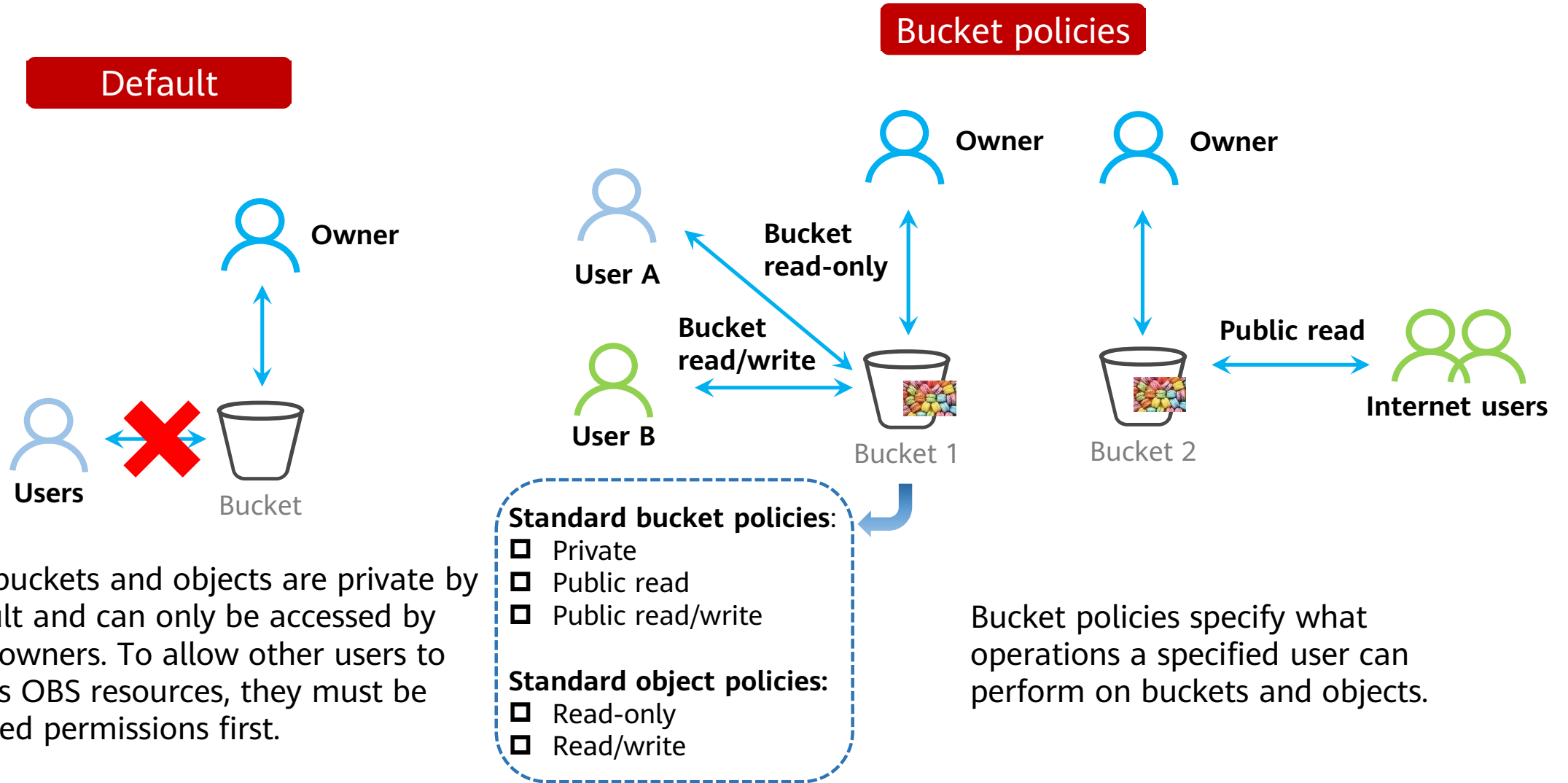
- Cross-region DR
- Regulatory compliance
- Cross-account replication not supported

Hosting a Website in OBS



To access website content, permissions must be set for objects and buckets.

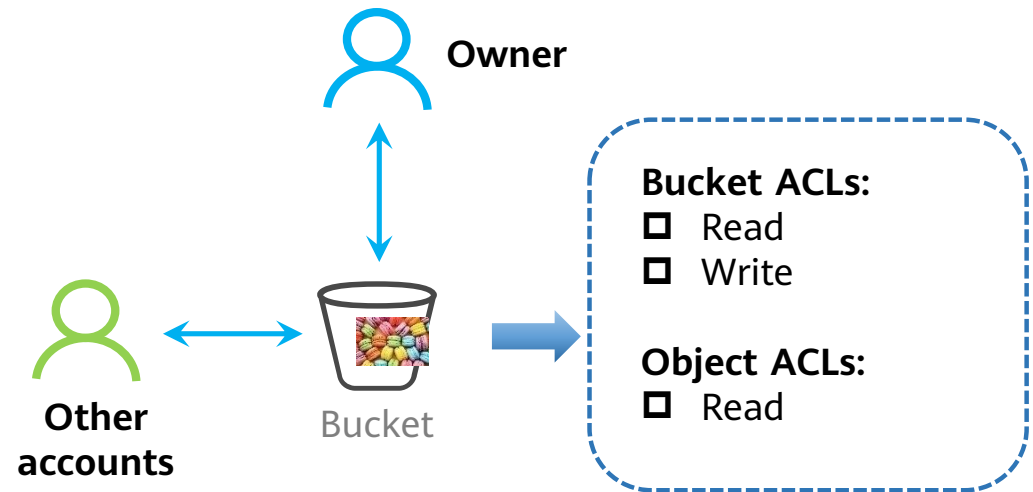
Granting Permissions in OBS – Bucket Policies



Granting Permissions in OBS – ACLs

ACLs

- ACLs stand for access control lists.
- ACLs grant account-level permissions on buckets and objects. By default, you can create an ACL when you create a bucket or object. Bucket or object owners have full control over their buckets or objects.
- To grant permissions beyond the account level, you need to use **IAM permissions or bucket policies**.



An Example Bucket Policy



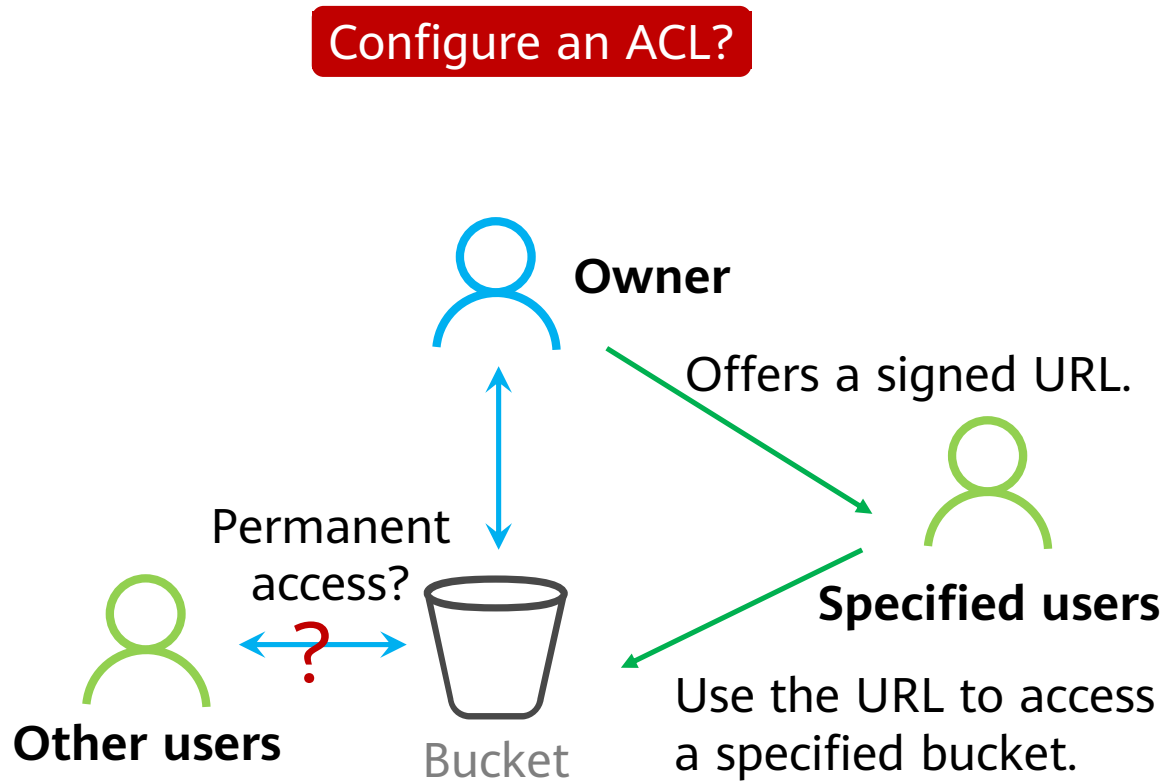
Bucket policy

```
{
  "Statement": [
    {
      "Sid": "static-site",
      "Effect": "Allow",
      "Principal": {
        "ID": ["*"]
      },
      "Action": [
        "List*",
        "Get*",
      ],
      "Resource": [
        "bkt-name",
        "bkt-name/*"
      ]
    }
  ]
}
```

A bucket policy statement consists of the following elements:

- ❑ Principals: Who can access resources
- ❑ Actions: What actions are allowed or denied
- ❑ Resources: Which resources can be accessed
- ❑ Conditions: When the policy is applied
- ❑ Effects: Whether the action is allowed or denied

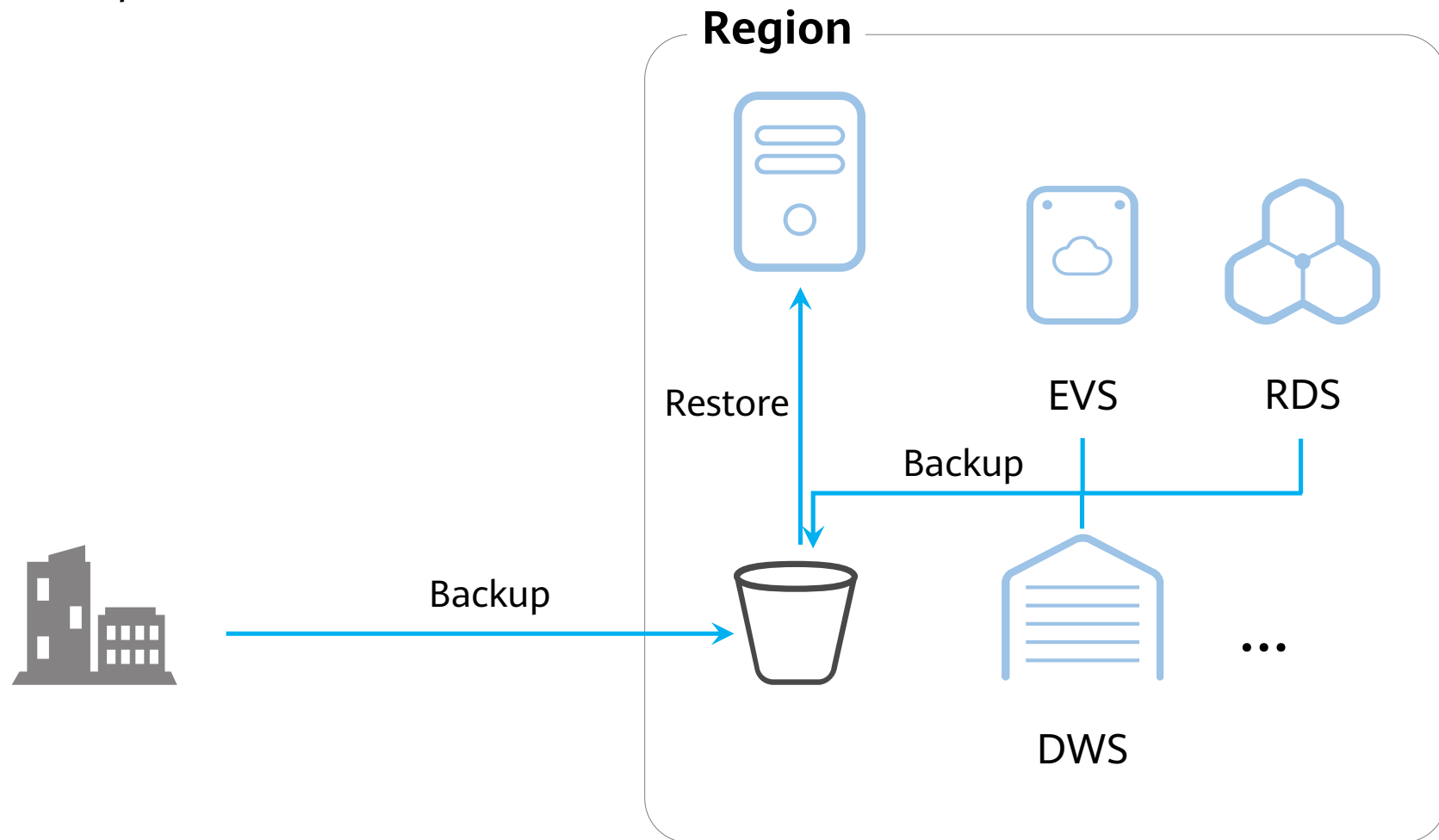
Signing a URL to Grant Temporary Access



- Objects are private.
- A URL needs to be signed to grant temporary access to other users.
 - ❑ Generally, a signed URL is created based on the code.
- A signed URL has a validity period.
- A signed URL is valid only when the AK/SK used for signature are valid.
- Specified users can use the provided URLs to perform specified operations within the validity period.
- Basic effects of URL validation

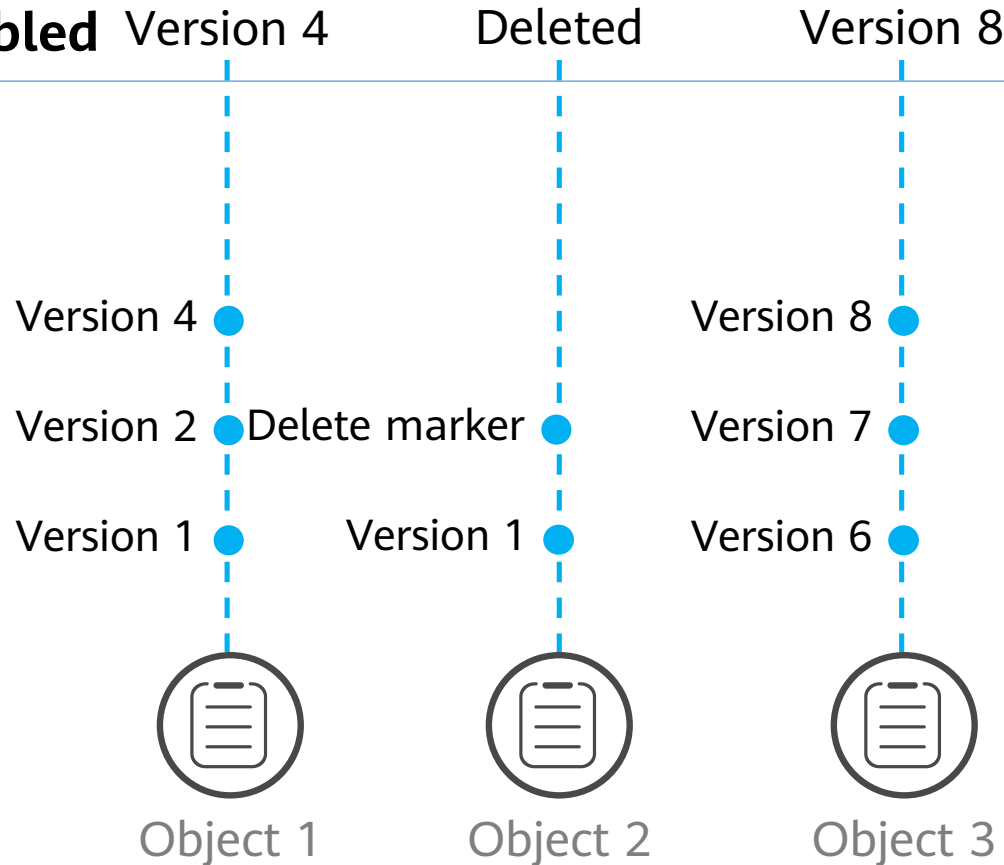
Storing Backups in OBS

You can use OBS to store data backups for EVS, RDS, and DWS.



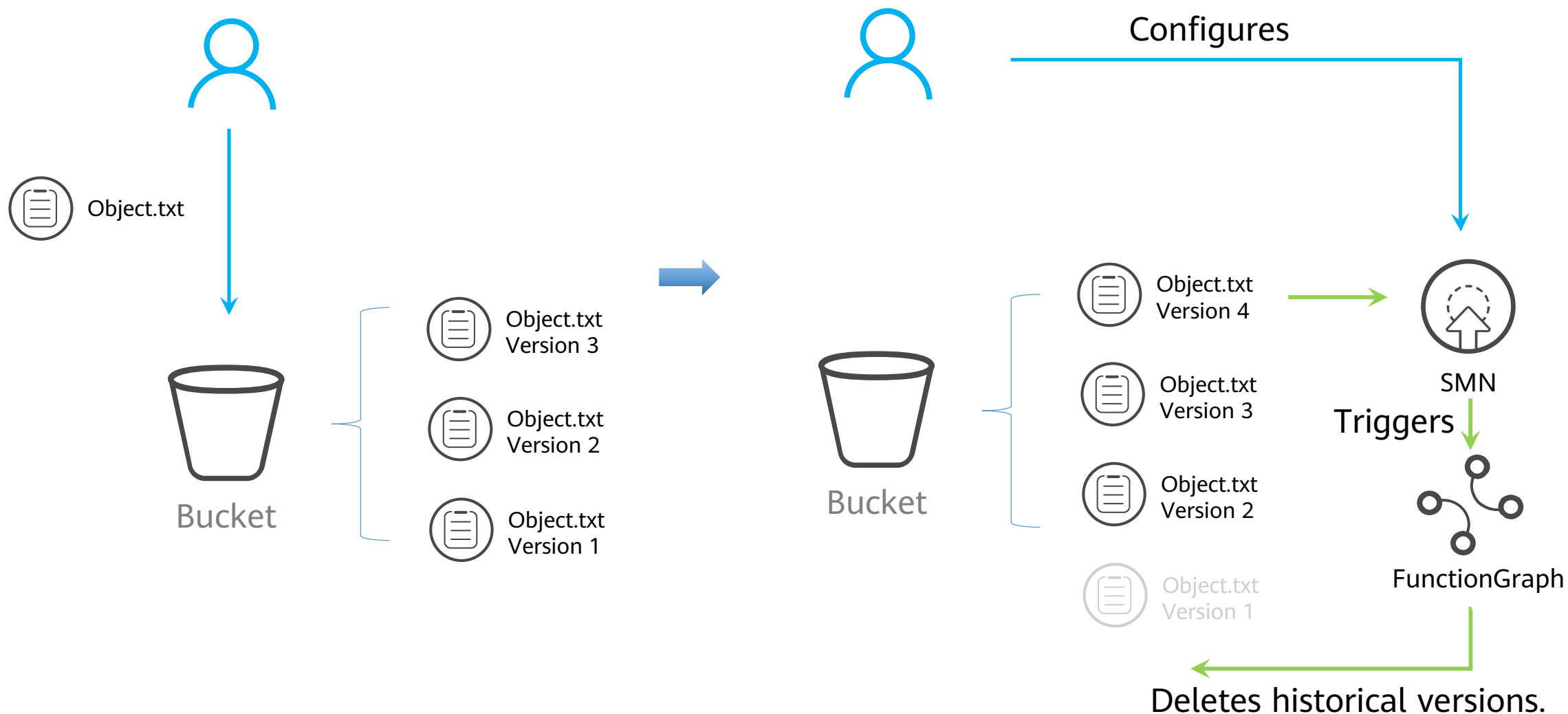
Managing Object Versions

Versioning disabled



- ❑ Versioning is disabled by default for new buckets in OBS. Existing objects will be overwritten by new ones.
- ❑ If versioning is enabled, OBS automatically creates a unique version ID for each uploaded object.
- ❑ When an object is deleted, OBS inserts a deletion marker to replace the object but this object can still be recovered.
- ❑ Storage for all object versions that do not have delete markers is billed.

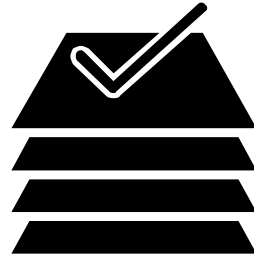
A Way to Delete Historical Object Versions



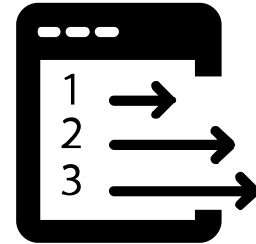
Benefits of Versioning



Accidental deletion protection



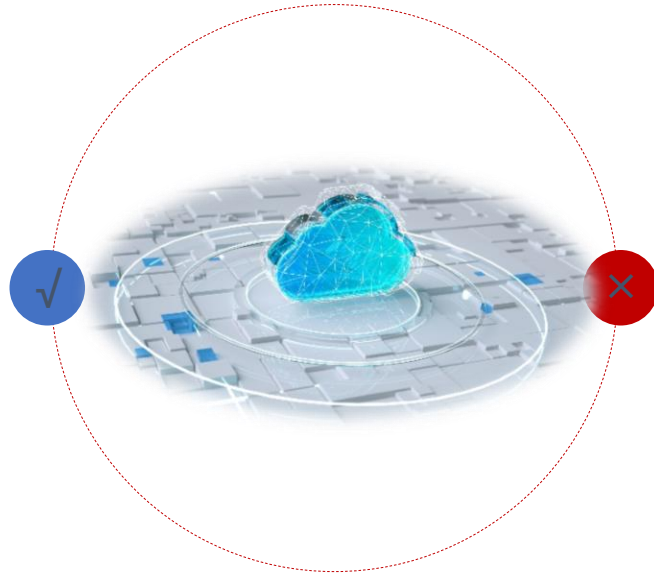
Multi-version backup



Release rollback

Use Cases for OBS Standard Storage

- ✓ Storage source for content distribution
- ✓ Write Once Read Many (WORM)
- ✓ No need to plan for storage capacity in the purchase
- ✓ Growing demands for storage capacity
- ✓ Compliance with regulatory or reliability requirements



- ✗ Data that needs to be changed frequently
- ✗ Existing code that is designed for accessing file systems
- ✗ Data transmission that requires ultra-low latencies
- ✗ Long-term archiving of data

OBS Infrequent Access Storage



OBS Infrequent Access

- Data that does not need to be accessed frequently, such as enterprise backups and active archives
- High reliability
- Real-time access
- Less expensive than OBS Standard storage

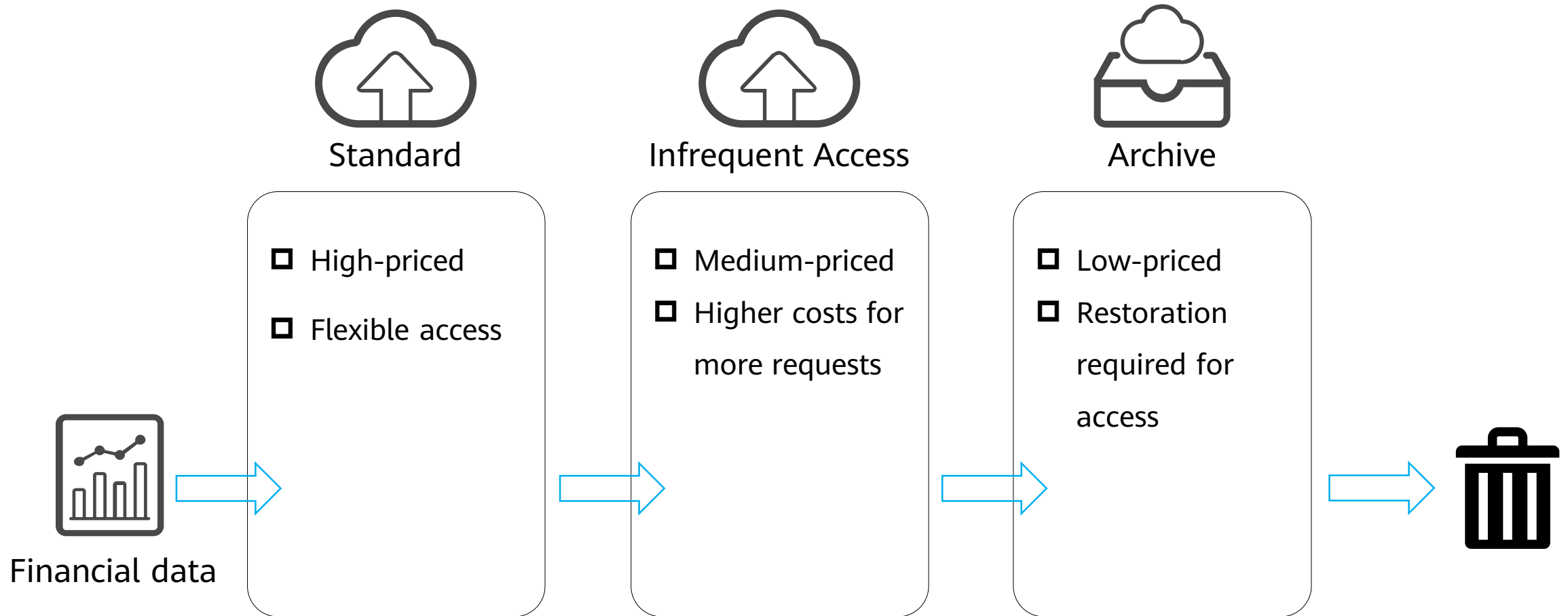
OBS Archive Storage



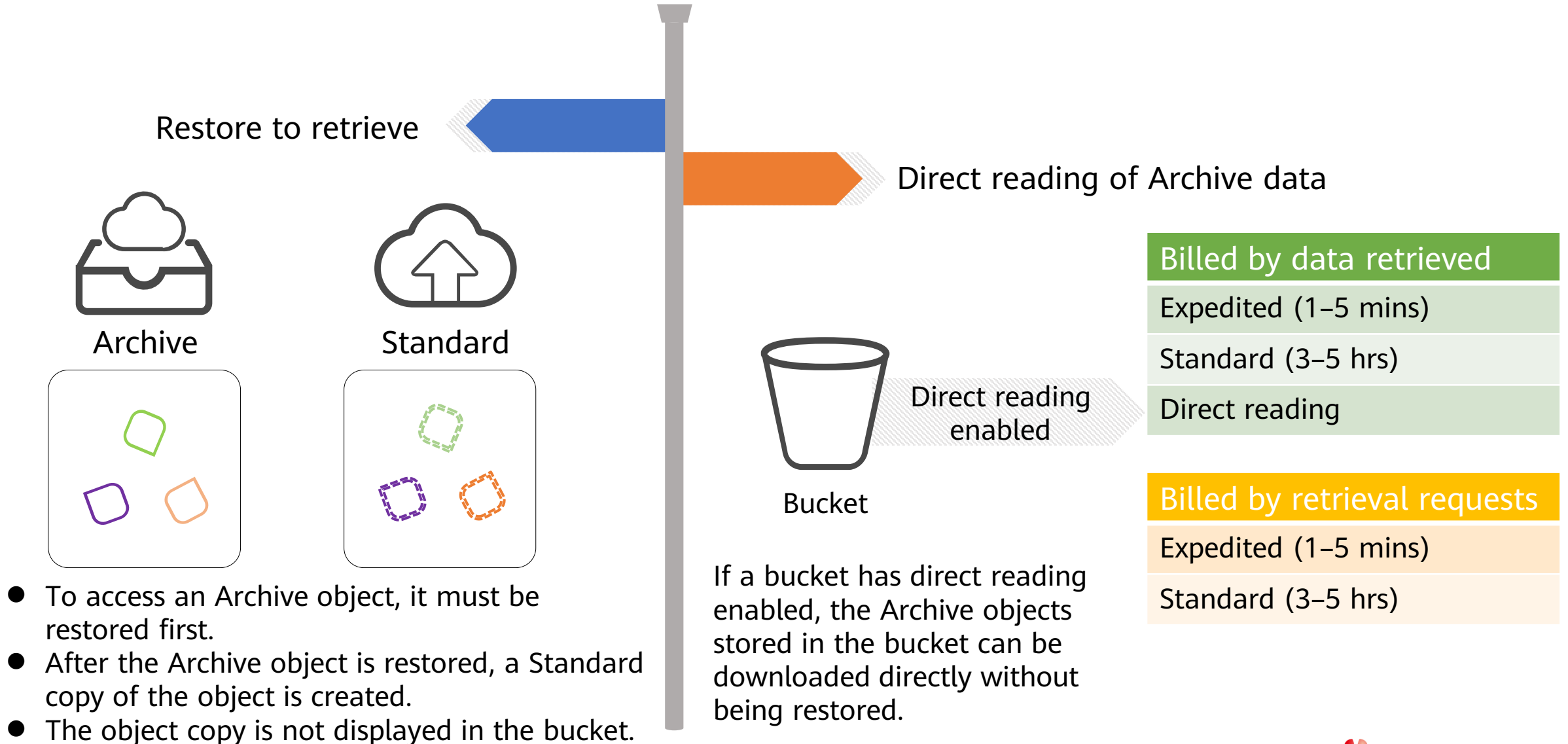
OBS Archive Storage

- Data that is seldom accessed but cannot be discarded
- High reliability
- Data must be restored before being retrieved.
- Less expensive than OBS Infrequent Access storage
- "Archive" is not the same as "backup".

Object Lifecycle Management (from Hot to Cloud)

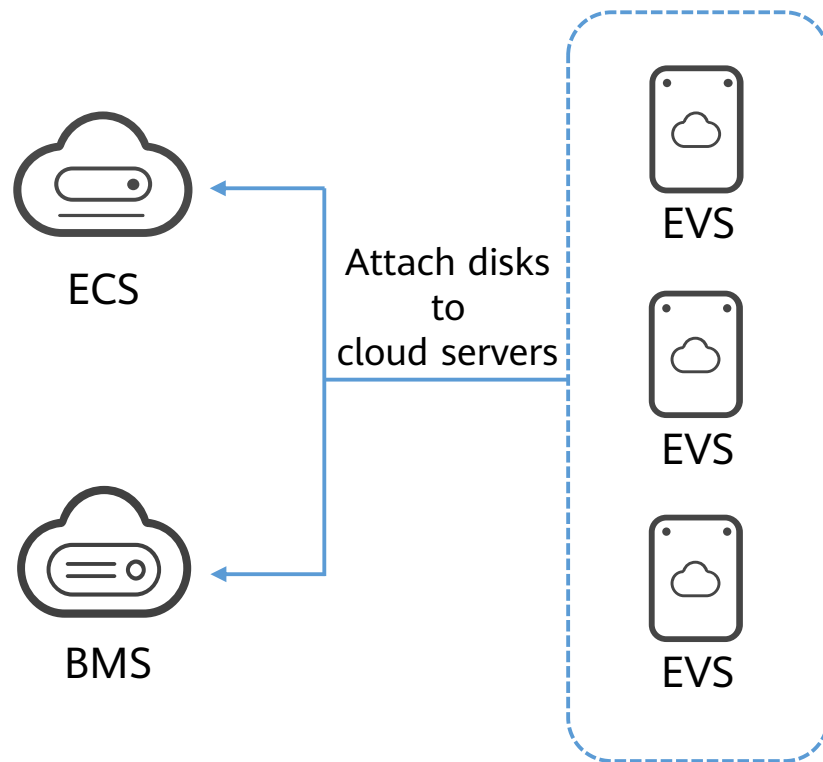


Two ways of Retrieving Archive Objects



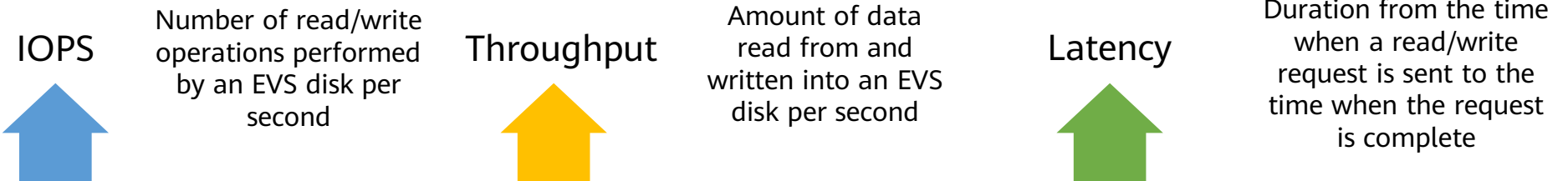
EVS

Elastic Volume Service (EVS) provides disks for Elastic Cloud Servers (ECSs), so it is a mandatory service.



- Single-AZ three-copy redundancy, 99.99999999% reliability
- Up to 32 TB per EVS disk
- Multiple specifications, offering high performance at reasonable prices
- Backup data stored to OBS

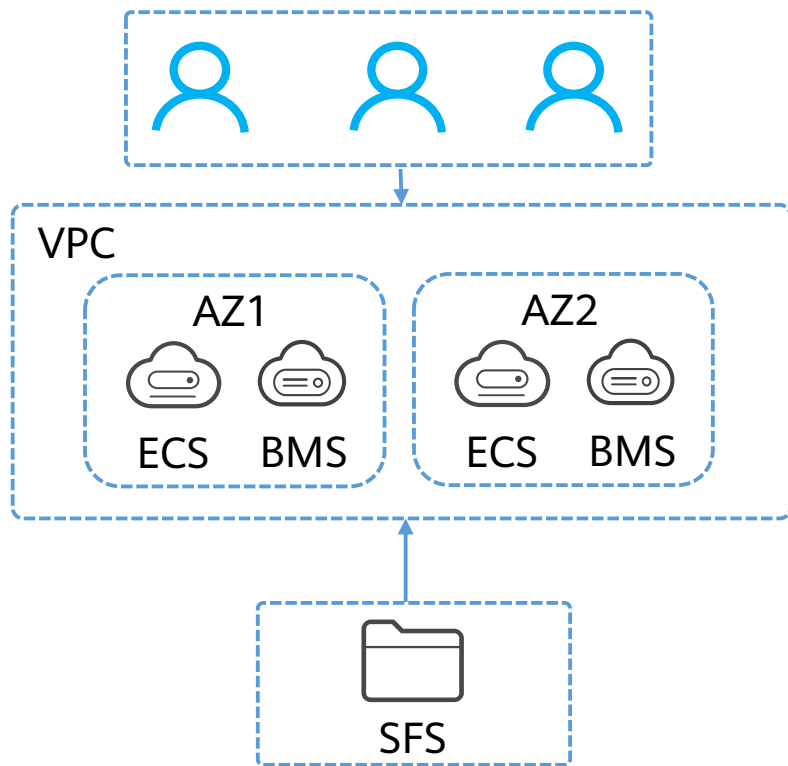
EVS Disk Type Selection



Parameter	Extreme SSD	Ultra-high I/O	General Purpose SSD	High I/O
Description	Superfast disks for workloads demanding ultra-high bandwidth and ultra-low latency	High performance disks excellent for enterprise mission-critical services as well as workloads demanding high throughput and low latency	Cost-effective disks designed for enterprise applications with medium performance requirements	Disks suitable for commonly accessed workloads
Max. IOPS	128,000	50,000	20,000	5,000
Max. throughput	1,000 MB/s	350 MB/s	250 MB/s	150 MB/s
Latency	200 μ s	1 ms	1 ms	1-3 ms
Cost	\$\$\$\$	\$\$\$	\$\$	\$
Typical use cases	Super-high IOPS and super-low latency, suitable for heavy-loaded databases and HPC file sharing	IOPS-optimized disks, suitable for I/O-intensive applications, high-performance system disks, and core file sharing	Cost-effective and highest throughput ratio, suitable for most workloads	Entry-level performance. With the burst performance, it can be used for workloads with unstable pressure, like boot disks.

SFS

Scalable File Service (SFS) provides reliable, high-performance shared file storage hosted on Huawei Cloud. With SFS, you can enjoy shared file access spanning multiple ECSs, Bare Metal Servers (BMSs), and containers created on Container Cloud Engine (CCE) and Cloud Container Instance (CCI).



- Support for mainstream NFS and CIFS protocols
- Multi-instance shared storage
- Automatic capacity scaling
- Linear performance increase
- Authorization-based security isolation

SFS Turbo File System Type Selection

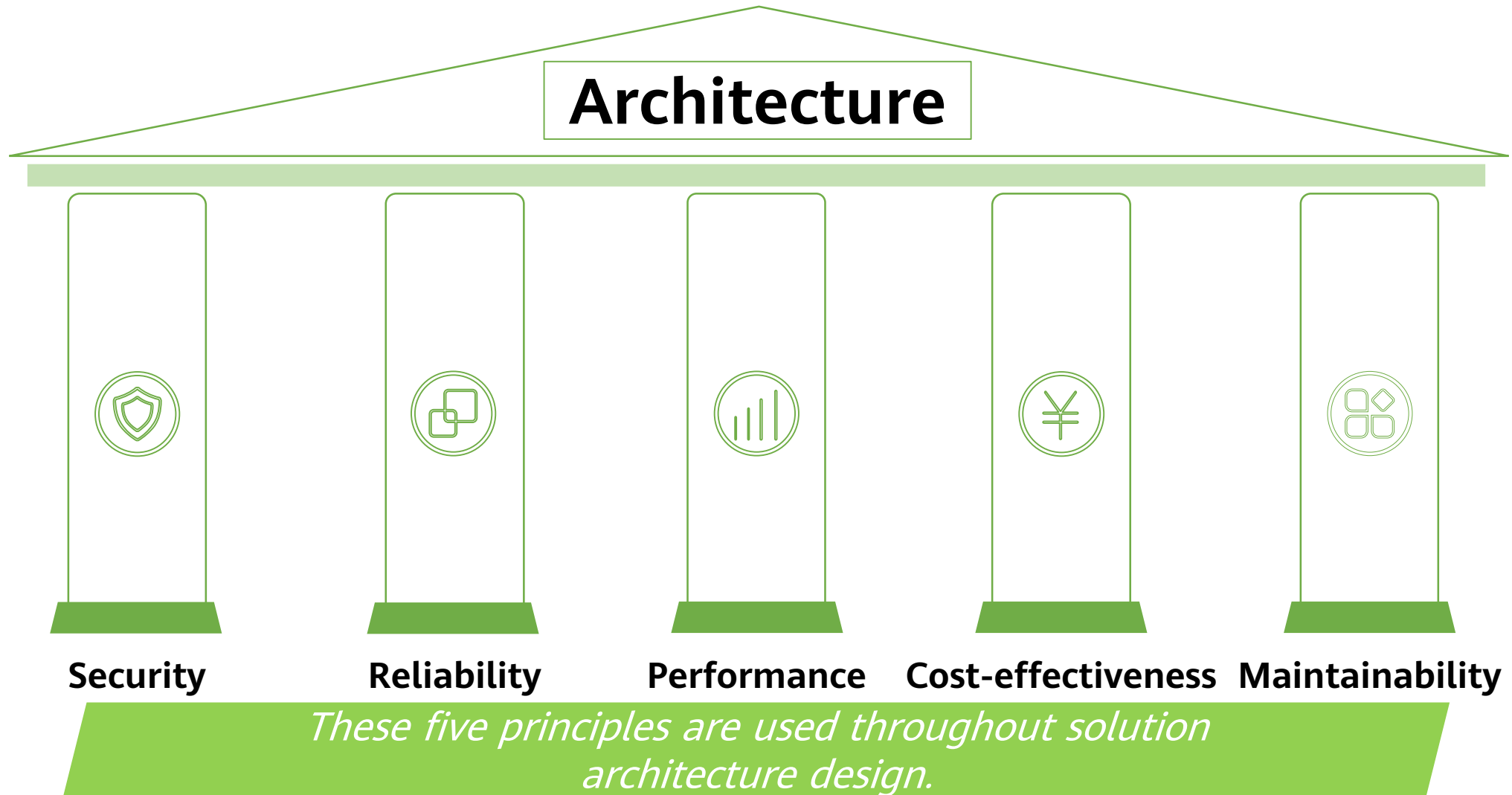
Parameter	General (Previous-generation File System)		HPC		
	SFS Turbo Standard	SFS Turbo Performance	40 MB/s/TiB	250 MB/s/TiB	1,000 MB/s/TiB
Max. bandwidth	150 MB/s	350 MB/s	8 GB/s	20 GB/s	20 GB/s Up to 200 GB/s
Max. IOPS	5,000	20,000	250,000	1 million	1 million
Latency	2-5 ms	1-2 ms	2-5 ms	< 1 ms	< 1 ms
Capacity	500 GB to 32 TB	500 GB to 32 TB	3.6 TB to 1 PB	1.2 TB to 1 PB	
Advantages	Large capacity and low cost	Low latency and high IOPS	Large capacity and low cost	Low latency and cost-effectiveness	High IOPS and high performance density
Typical use cases	Code storage, log storage, file sharing, and enterprise OA	High-traffic websites, file sharing, content management, image rendering, AI training, and enterprise OA	Enterprise OA and code repository management	AI training, EDA simulation, gene analysis, and video rendering	Autonomous driving, AI generated content, and EDA in chip design



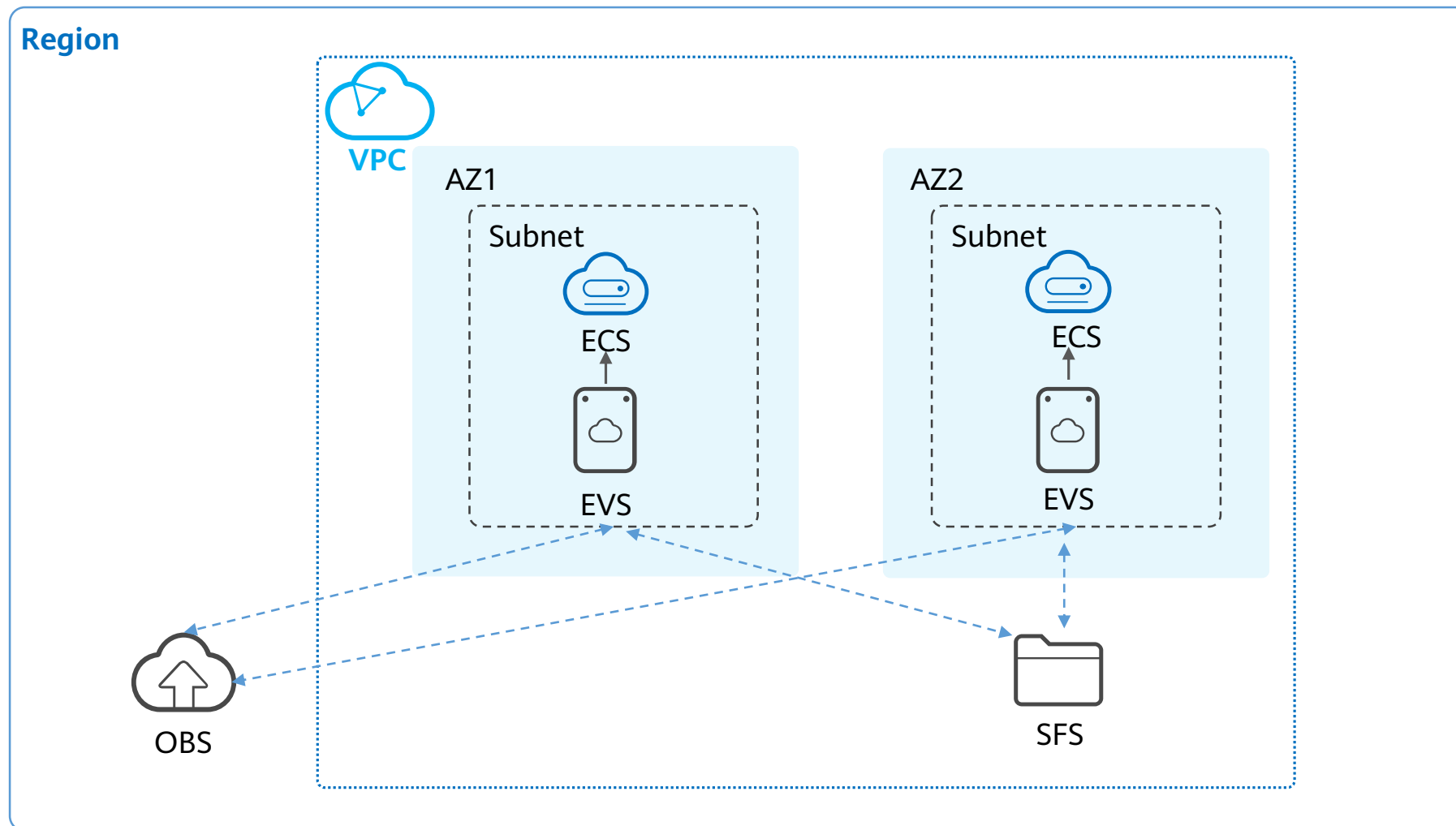
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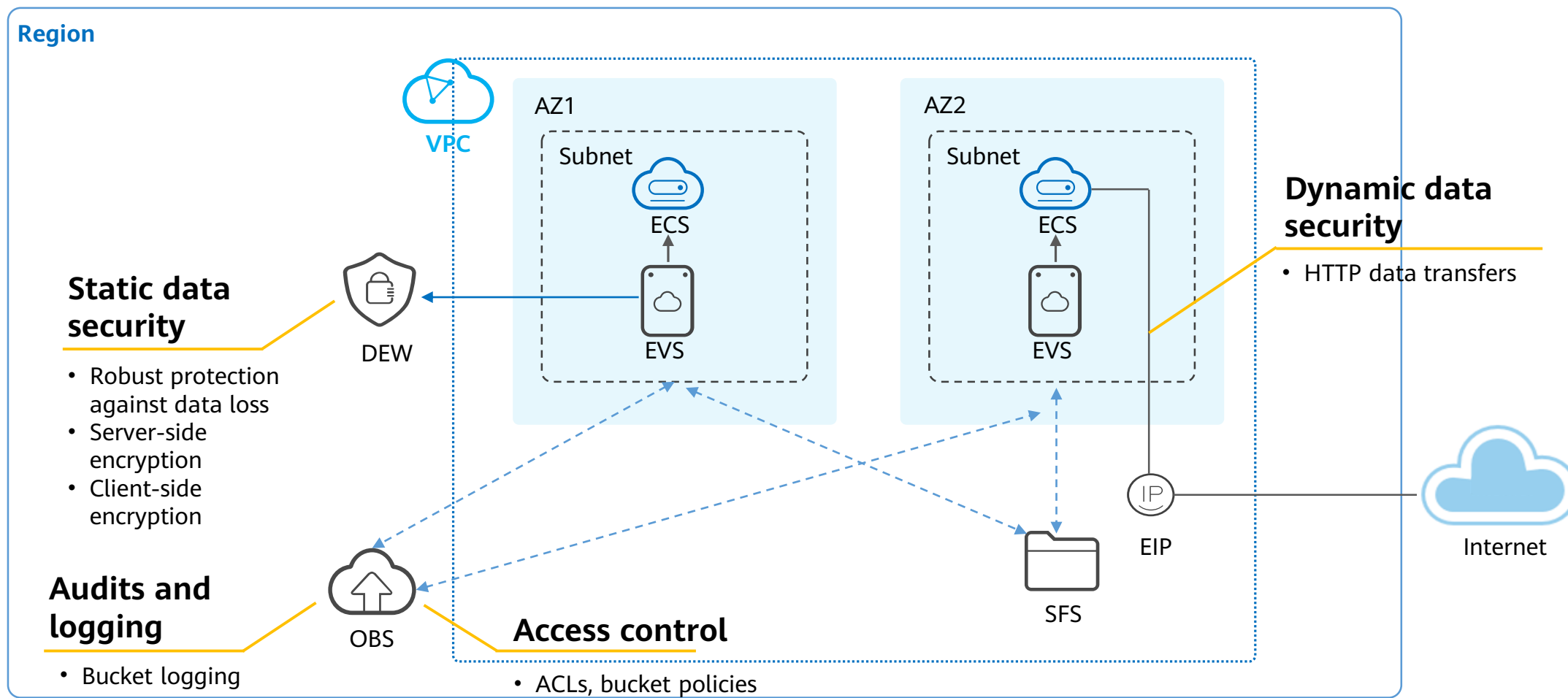
Five Principles of Storage Solution Design



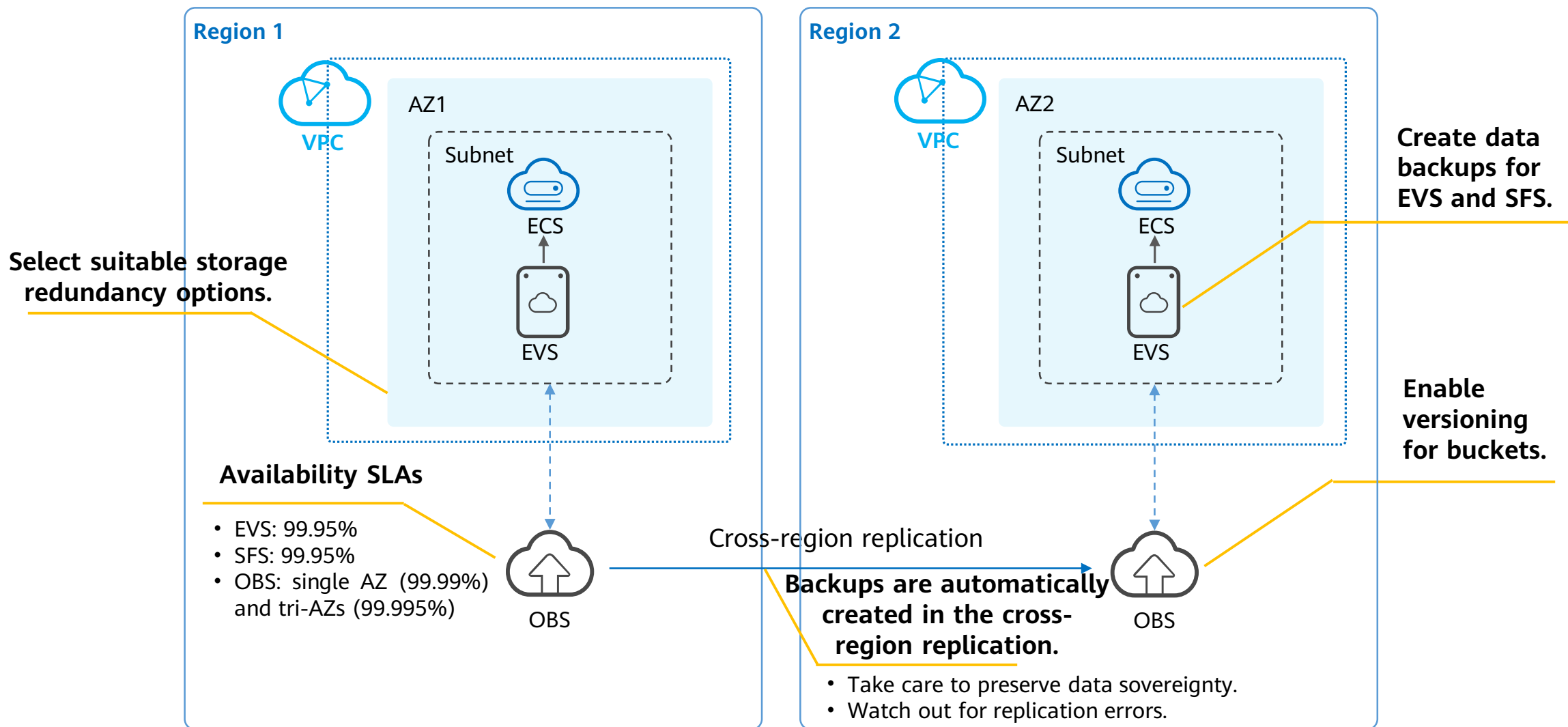
An Example Architecture



Storage Security



Storage Reliability



Storage Performance of OBS

- » Latency as low as 10 ms
- » High concurrency and bandwidth
- » You need to avoid hot partitions.

You are advised to use random prefixes for object names.

```
.....  
waasmedic.20210720_061549_205.etl  
waasmedic.20210720_063505_217.etl  
waasmedic.20210720_071052_011.etl  
waasmedic.20210720_072451_413.etl  
waasmedic.20210720_072854_473.etl  
waasmedic.20210720_082109_327.etl
```

```
WindowsUpdate.20210720.115158.03  
4.6.etl  
WindowsUpdate.20210720.115158.03  
4.7.etl  
WindowsUpdate.20210720.115158.03  
4.8.etl  
WindowsUpdate.20210720.123237.68  
9.1.etl  
WindowsUpdate.20210720.131930.89  
9.1.etl  
.....
```

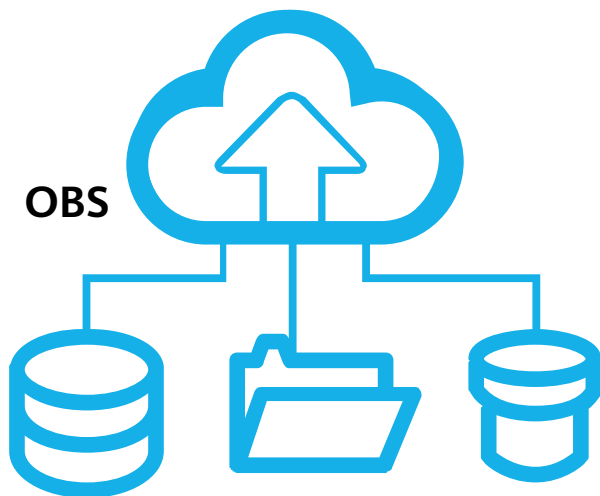
```
.....  
268a.WindowsUpdate.20210720.123237.etl
```

```
7cef.WindowsUpdate.20210720.131930.etl  
7da3.IExplorer.20210611.081646.etl  
.....  
7f1d.SecEssentials.20210226.112317.etl
```

```
a133.waasmedic.20210720_082109_327.etl  
.....
```

Storage Cost-effectiveness of OBS

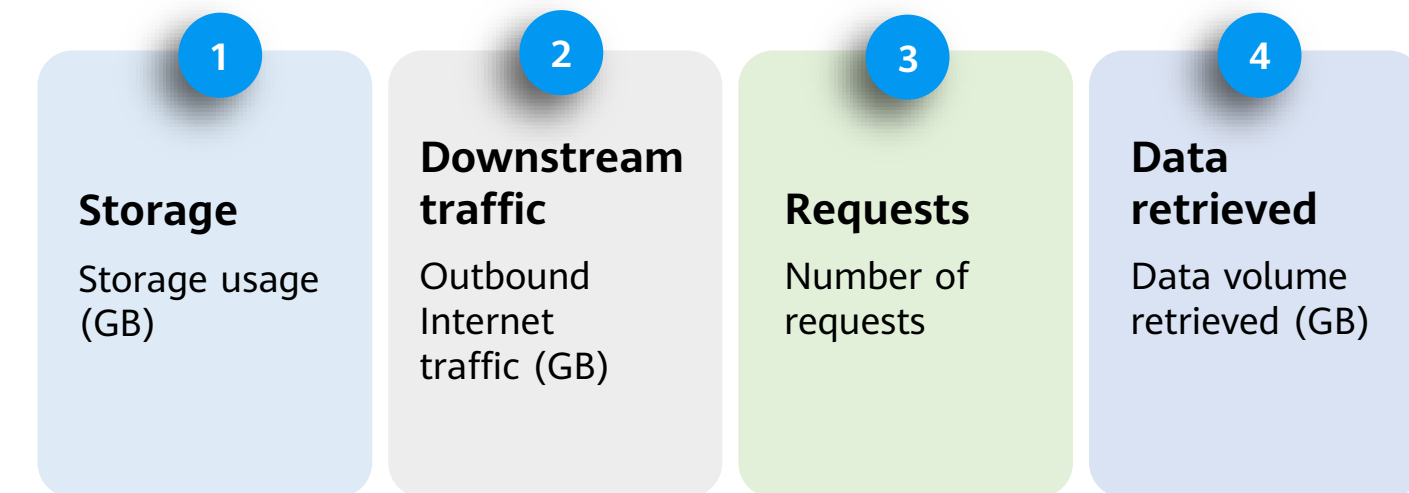
» Decoupled storage and compute for low storage costs



Standard Infrequent Access Archive

» Select a proper storage class to reduce costs.

- Do not select Infrequent Access to store hot data.
- Do not select Archive if you need to retrieve data immediately when need.



» OBS has four billing items: storage, traffic, requests, and data retrieval.

- Inbound traffic and the traffic transferred out to the same region are free.
- Discounts on traffic from OBS to CDN
- Avoid unnecessary function calls to reduce costs.

Storage Maintainability of OBS

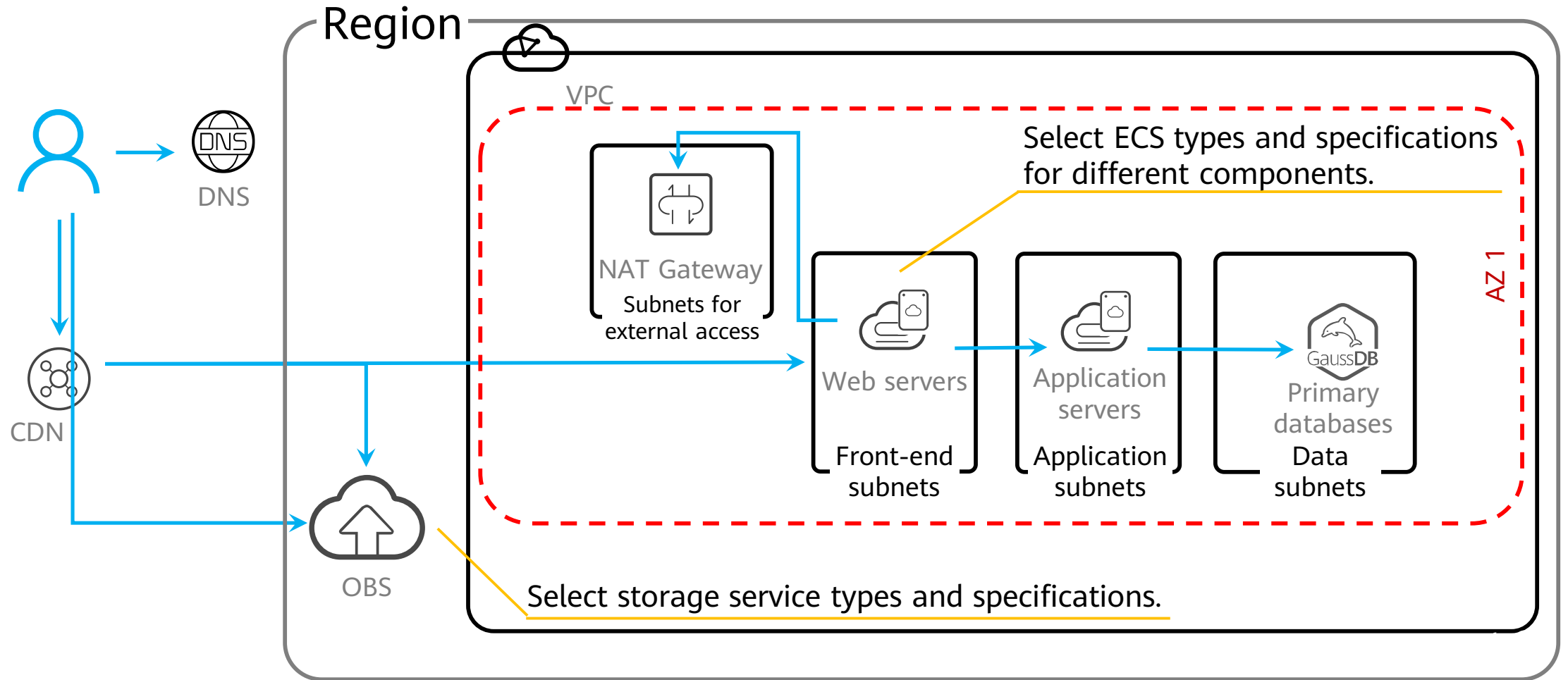
- Fully managed services require little maintenance.
- When versioning is enabled, historical versions need to be properly managed.
- Access record analysis and permissions management



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Compute and Storage Service Practices



Quiz

(Single-choice question) A company found in an annual audit that their OBS resources were over-budget. What can the company do to reduce costs?

- A. Disable versioning for buckets.
- B. Optimize access policies to grant only required permissions.
- C. Configure lifecycle policies to transition objects to cost-effective storage classes.
- D. Enable direct reading for buckets.

Summary

This lesson presented an overview of Huawei Cloud storage solutions. Basic cloud storage services (OBS, SFS, and EVS) were introduced. OBS features such as reliability, cross-region replication, permission control, and lifecycle management were described. The five principles of designing storage solutions were then explained, followed by the compute and storage service practices.



Acronyms and Abbreviations

- EVS: Elastic Volume Service
- OBS: Object Storage Service
- SFS: Scalable File Service
- IOPS: Input/Output Operations per Second
- NAS: Network Attached Storage
- URL: Uniform Resource Locator
- CCE: Cloud Container Engine
- CCI: Cloud Container Instance

Thank You.

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